Program:5

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Problem Statement:

Implement HPC application for AI/ML domain

In [1]:

```
import pandas as pd
import re
import nltk
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from joblib import Parallel , delayed
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
```

```
C:\Users\dell\anaconda3\Lib\site-packages\pandas\core\computation\expressio ns.p
y:22: UserWarning: Pandas requires version '2.10.2' or newer of 'numexpr' (vers ion '2.8.7' currently
installed). from pandas.core.computation.check import NUMEXPR_INSTALLED
C:\Users\dell\anaconda3\Lib\site-packages\pandas\core\arrays\masked.py:56: User Warning:
Pandas requires version '1.4.2' or newer of 'bottleneck' (version '1.3.7' currently installed). from
pandas.core import (
```

In [3]:

```
nltk.download('stopwords') nltk.download('wordnet')
```

```
[nltk_data] Downloading package stopwords to
[nltk_data]  C:\Users\dell\AppData\Roaming\nltk_data...
[nltk_data]  Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data]  C:\Users\dell\AppData\Roaming\nltk_data... [nltk_data]  Package wordnet is
already up-to-date!
```

Out[3]: True

In [5]:

```
# load dataset
df = pd.read_pickle("News_dataset.pickle")

texts = df['Content'].astype(str).tolist() labels =
df['Category'].tolist()
```

```
# preprocessing setup
stop_words = set(stopwords.words('english'))
```

In [7]:

```
lemmatizer = WordNetLemmatizer()
```

In [9]:

```
# function def
preprocess(text):
    text = text.lower()
    text = re.sub(r'[^a-z\s]', '', text) words = text.split() words =
    [w for w in words if w not in stop_words] words =
    [lemmatizer.lemmatize(w) for w in words] return "
    ".join(words)
```

In [13]:

```
# HPC (parallel processing using all CPU cores)
clean_texts = Parallel(n_jobs=-1)( delayed(preprocess)(t) for
t in texts
)
```

In [15]:

```
# TF-IDF tfidf = TfidfVectorizer(max_features=1000)
X = tfidf.fit_transform(clean_texts)
```

In [17]:

```
# ML model
model = LogisticRegression()
model.fit(X, labels)
```

Out[17]:

```
LogisticRegression
LogisticRegression()
```

In [21]:

```
print("HPC ML Task Completed ")
```

```
HPC ML Task Completed
```

In [ ]: